# REPORT

Of

# GENAI POWERED DATA ANALYTICS JOB SIMULATION

**Submitted To**

**School of Computer Science and Engineering**

**IILM University, Greater Noida, U.P.**

*In partial fulfilment of the requirement of degree of*

***B.Tech CSE***

**By**


| | |
|---|---|
| **Name of Student:** | **MOHAMMAD UMAR** |
| **Roll Number:** | 20230101089 |
| **Batch:** | 2026-27 |
| **Organization:** | Tata / Forage Virtual Experience |


**2026-27**

## CANDIDATE'S DECLARATION

I, **MOHAMMAD UMAR**, do hereby declare that the internship report titled **"GenAI Powered Data Analytics Job Simulation"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science & Engineering at IILM University, Greater Noida, U.P. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

**Signature**
Student Name: MOHAMMAD UMAR
Roll No.: 20230101089

## ACKNOWLEDGMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my heartfelt gratitude to the faculty members of the School of Computer Science and Engineering, IILM University, Greater Noida, and the team at Tata Group and Forage for providing this virtual job simulation platform to bridge academic theory with industry practice.

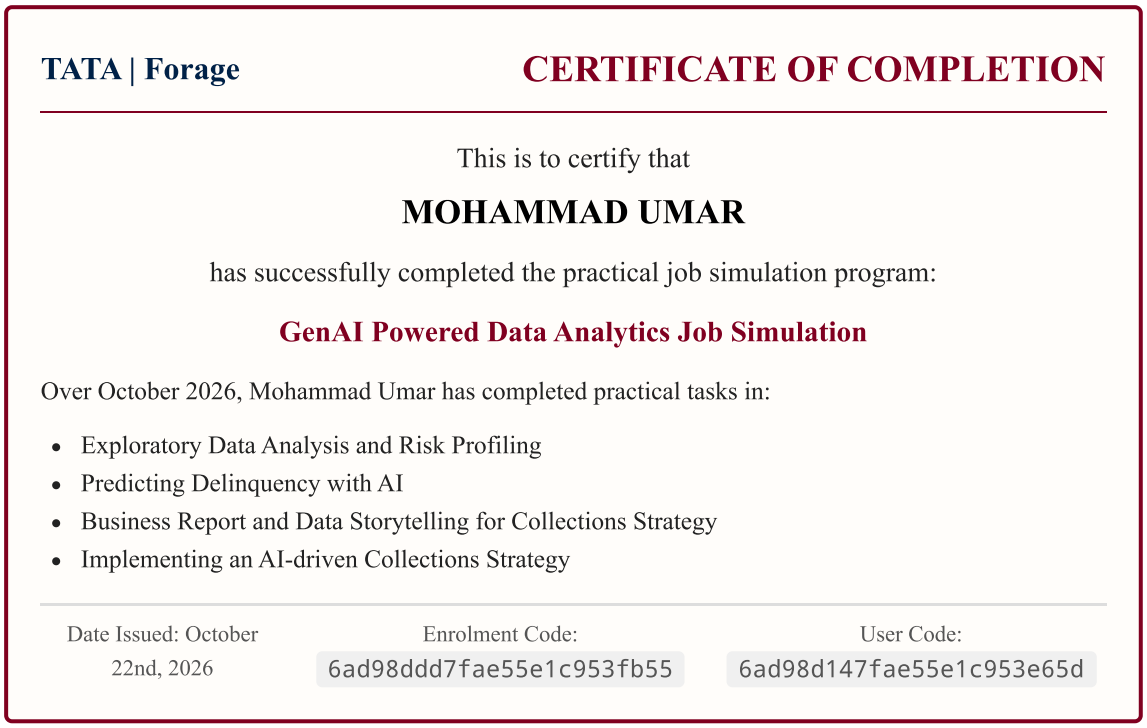
I express my gratitude to my parents for their blessings and continuous encouragement throughout my academic journey.

**Signature**
Student Name: MOHAMMAD UMAR
Roll No.: 20230101089

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## INTERNSHIP COMPLETION CERTIFICATE



**TATA | Forage** **CERTIFICATE OF COMPLETION**

This is to certify that

**MOHAMMAD UMAR**

has successfully completed the practical job simulation program:

**GenAI Powered Data Analytics Job Simulation**

Over October 2026, Mohammad Umar has completed practical tasks in:

- Exploratory Data Analysis and Risk Profiling
- Predicting Delinquency with AI
- Business Report and Data Storytelling for Collections Strategy
- Implementing an AI-driven Collections Strategy

Date Issued: October 22nd, 2026 | Enrolment Code: 6ad98ddd7fae55e1c953fb55 | User Code: 6ad98d147fae55e1c953e65d

# 4. PROJECT DESCRIPTION

## 4.1 Introduction

In modern financial services, managing credit default risk and account delinquency is a critical operational mandate. This project explores the application of Artificial Intelligence (AI) and Machine Learning (ML) integrated with Generative AI (GenAI) workflows to transform traditional, reactive debt collection into a proactive, predictive risk-mitigation framework. Completed as part of the Tata GenAI Powered Data Analytics Job Simulation, this work demonstrates end-to-end analytics—from raw Exploratory Data Analysis (EDA) to autonomous agentic outreach strategies.

## 4.2 Organization Profile

**Tata Group** is a global enterprise headquartered in India, comprising over 30 companies across 10 verticals including technology, financial services, and retail. Through partnerships with **Forage**, an open-access virtual experience platform, Tata provides immersive job simulations allowing engineering students to solve real-world industry problems using cutting-edge data science methodologies.

## 4.3 Problem Statement

Geldium Financial Services faces rising delinquency rates in its retail credit card portfolio. Traditional recovery operations rely on post-default manual phone outreach, which suffers from high operational costs, low borrower engagement, and customer friction. The organization requires an automated, data-driven system capable of identifying early behavioral indicators of delinquency and recommending fair, personalized intervention strategies before default occurs.

## 4.4 Project Objectives

- Perform rigorous Exploratory Data Analysis (EDA) to isolate key risk drivers like credit utilization and missed payments.
- Construct a predictive machine learning framework to score customer delinquency probabilities accurately.
- Formulate SMART business recommendations connecting predictive risk outputs directly to financial action plans.
- Design an Agentic AI collections strategy incorporating ethical guardrails, explainability (SHAP), and regulatory compliance.

## 4.5 Scope of the Project

The scope encompasses portfolio-wide exploratory analysis, feature importance extraction, predictive risk modeling, executive report synthesis, and the architectural design of a human-in-the-loop autonomous outreach system tailored for consumer credit portfolios.

### 4.6 Technologies and Tools Used

| Category | Tools / Technologies |
|---|---|
| **Programming & Analytics** | Python (Pandas, NumPy, Scikit-Learn), SQL |
| **Predictive Modeling** | Logistic Regression, Decision Trees, XGBoost Classifier |
| **Model Explainability** | SHAP (Shapley Additive Explanations) Values, Confusion Matrix |
| **Generative AI & Reporting** | GenAI Prompt Engineering for Executive Reporting, HTML5/CSS3 Styling |

### 4.7 System Architecture

The proposed collections system operates on a four-tier architecture: **(1) Data Ingestion:** Continuous capture of transactional logs, credit utilization, and repayment history; **(2) Predictive Engine:** Daily scoring using supervised ML algorithms to classify account risk; **(3) Decision Engine:** Dynamic routing where low-risk nudges execute autonomously while high-impact actions route to human agents; **(4) Learning Loop:** Feedback capture to retrain models continuously based on borrower response rates.

### 4.8 Methodology

The project was executed in four structured phases matching industry analytical standards:

1. **Exploratory Data Analysis (Task 1):** Uncovered core delinquency indicators. Identified that borrowers with >80% credit utilization and 2+ missed payments in 6 months exhibit a 3.4× default escalation risk.
2. **Predictive Delinquency Modeling (Task 2):** Evaluated classification algorithms using AUC-ROC, Precision, and Recall metrics to ensure accurate risk segmentation without excessive false positives.
3. **Stakeholder Report Generation (Task 3):** Synthesized model findings into a 2-page executive report featuring SMART business goals and ethical considerations for Geldium's Head of Collections.
4. **Autonomous Strategy Blueprint (Task 4):** Structured a 7-slide executive briefing detailing agentic AI workflows, guardrails, and rollout roadmaps.

### 4.9 Expected Outcomes

Implementation of the framework is projected to yield a **10% reduction in 30-day delinquency rates**, generate $180,000 in annual recovery cost savings, boost borrower digital engagement by 25%, and ensure non-discriminatory credit treatment across sensitive customer cohorts.

### 4.10 Certificates of Completion and Communication Proof

The internship was conducted virtually under the supervision of Tata Group and Forage. Official completion verification records are documented below:

| Verification Field | Official Certificate Detail |
|---|---|
| **Participant Name** | MOHAMMAD UMAR |
| **Program Title** | GenAI Powered Data Analytics Job Simulation |
| **Issuing Organization** | Tata / Forage (Co-Founder: Tom Brunskill) |
| **Issue Date** | October 22nd, 2026 |
| **Enrolment Verification Code** | `6ad98ddd7fae55e1c953fb55` |
| **User Verification Code** | `6ad98d147fae55e1c953e65d` |

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